

# LifeAtlas Intern Ecosystem Overview

Cross-Stream Scope, Goals, and Challenges - Phase 2/3 Transition  
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1. Boardy (AI Matchmaking)

Phase 2/3

Core Problem:

Cold outreach & LinkedIn networking are broken. Boardy replaces them with a 5-minute automated voice conversation and intelligent matching to connect founders with relevant network contacts.

Current Focus:

Building a custom voice pipeline using Pipecat (Silero VAD, AssemblyAI STT, OpenRouter LLM, Cartesia TTS) to replace Retell AI and reduce per-call API cost from ~\$0.19 to ~\$0.02. Replacing Voyage/Claude with local models.

Tech Stack:

Pipecat, Python, OpenAI/OpenRouter APIs, Cartesia TTS, local BGE/Ollama models.

4. Security & Auditing (ZeroClaw)

Phase 2 -> 3

Core Problem:

Security risks across active production codebases and container boundaries. Need to detect credential leaks, unsafe injection patterns, and verify tenant isolation boundaries in Supabase.

Current Focus:

Fully integrated Secret and Code Pattern scanners. Setting up automated test suites for Auth Bypass, Session Management, and Rate Limiting on FastAPI. Auditing the Stream 3 digital twin container sandbox.

Tech Stack:

FastAPI, Supabase RLS, Python (ZeroClaw Scanner), Docker, CI/CD Gates.

2. DataPro+ (Datacenter Deals)

Phase 0 (70%)

Core Problem:

Gated and complex datacenter investment deals that require intensive role-based NDA workflows, manual referral tracking, and lead management for land, power, and infrastructure opportunities.

Current Focus:

Building a secure dealflow platform featuring AI-based PDF extraction for documents, modular contract blocks (for identity/NDAs), and a data migration framework. Monetized through 1-3% deal fees.

Tech Stack:

React, Vite, Tailwind CSS, FastAPI, Supabase, Gemini AI.

5. LPI (Life Goals & SMILE)

Phase 2 -> 3

Core Problem:

Helping users plan, structure, and track progress on life goals using the SMILE methodology, which requires complex activity ingestion, scoring algorithms, and recommendations.

Current Focus:

Developing Supabase persistence layers, establishing dual-sync logging, resolving database silent migration bugs, connecting React frontend components to FastAPI endpoints, and creating activity pipelines.

Tech Stack:

FastAPI, Supabase, React, Python, MCP servers, mypy, ruff.

3. VSAB (Production Digital Twin)

Phase 2

Core Problem:

Lack of real-time factory floor monitoring and production optimization for VSAB Gruppen (Swedish manufacturer). Needs automated risk detection with penalty formulas for delay tracking.

Current Focus:

Split in two teams: Team A is building a Streamlit dashboard with an integrated AI assistant, animated sketch views, and product comparison metrics. Team B is building a 3D WebGL factory simulation with layout migration.

Tech Stack:

Streamlit, Three.js, WebGL, D3, Recharts, Pandas, Python.

6. MRO Optimization (Our Stream)

Phase 2 -> 3

Core Problem:

Optimizing Altiostar's Tokyo 5G network handovers. Need to dynamically adjust Cell Individual Offsets (CIO) across 763 neighbor relations to keep call drops low under environmental stress.

Current Focus:

Trained PPO RL models on relation-level PM data for 50k steps. Built an automated Ship Gate Validator to check production thresholds (HO Success >99%, Ping-Pong <5%) and created a policy ONNX exporter.

Tech Stack:

Python, Stable-Baselines3, PPO, PyTorch, ONNX, Streamlit, MLflow.